

Spread Locator: A Statistical Distribution Analysis Model

◆ Part A - Theoretical Foundation

Q1. What is Statistical Distribution?

Ans: A statistical Distribution describes how the values of a variable are spread or distributed across different possible outcomes. It shows the probability of occurrence of each value in a dataset.

Example:

If we roll a fair die, the numbers 1,2,3,4,5 and 6 each have a probability of $\frac{1}{6}$.

Uses:

- Understanding data patterns
- Making predictions
- Statistical analysis and decision-making

Q2. What is Q-Q Plot and why it is used ?

Ans: Q-Q (Quantile-Quantile) Plot is a graphical method used to compare the distribution of a dataset with a theoretical distribution.

Purpose:

- Check whether data follows a normal distribution.
- Identify outliers and skewness.

Interpretation:

- Points on a straight line → Data is normally distributed.

- Points away from the line → Data is not normally distributed.

Q3. Difference Between Discrete and Continuous Distributions

Ans:

Discrete Distribution	Continuous Distribution
Takes countable values	Takes infinite values within a range
Examples: Number of students, dice outcomes	Examples: Height, Weight, Temperature
Uses Probability Mass Function (PMF)	Uses Probability Density Function (PDF)

Example:

- Discrete: Number of heads in 10 coin tosses.
- Continuous: Time required to complete a task.

Q4. What is Bernoulli Distribution?

Ans: A Bernoulli Distribution models an experiment with only two possible outcomes.

- Success(1)
- Failure(0)

Probability:

- $P(X=1) = p$
- $P(X=0) = 1-p$

Example:

Tossing a coin:

- Head = Success
- Tail = Failure

Mean:

$$\mu = p$$

Variance:

$$\sigma^2 = p(1-p)$$

Q5. What is Binomial Distribution?

Ans: Binomial Distribution represents the number of successes in a fixed number of independent Bernoulli trials.

Conditions:

1. Fixed number of trials(n)
2. Two outcomes per trial
3. Constant probability(p)
4. Independent trials

Formula:

Example:

Probability of getting exactly 3 heads in 5 coin tosses.

Mean:

$$\mu = np$$

Variance:

$$\sigma^2 = np(1-p)$$

Q6. Explain Log - Normal Distribution

Ans: A variable follows a Log-Normal Distribution if its logarithm is normally distributed.

Characteristics:

- Values are always positive.
- Right-skewed distribution.

Example:

- Income distribution
- Stock prices
- Biological measurements

Applications:

Used when data grows multiplicatively rather than additively.

Q7. Explain Power Law Distribution

Ans: A Power Law Distribution occurs when large events are rare but small events are very common.

Formula:

$$P(X=x)=Cx^{-\alpha}$$

Where:

- C = Constant
- α = Power law exponent

Examples:

- City populations
- Earthquake magnitudes
- Social media followers

Characteristic:

- Heavy tails
- Extreme values occur more frequently than expected

Q8. What is Box-Cox Transform?

Ans: The Box-Cox Transform is a mathematical transformation used to make data more normally distributed.

Used:

- Reduce skewness
- Improve normality
- Enhance statistical model performance

Example:

Transforming highly skewed income data before regression analysis.

Q9. Explain Poisson Distribution with an Example

Ans: A Poisson Distribution models the number of times an event occurs within a fixed interval of time or space.

Example:

If a cell center receives 5 calls per hour on average, the number of calls received in the next hour follows a Poisson Distribution.

Mean:

$$\mu = \lambda$$

Variance:

$$\sigma^2 = \lambda$$

Applications:

- Website visits
- Traffic accidents
- Customer arrivals

Q10. What is Z-Score Probability?

Ans: A Z-Score measures how many standard deviations a value is away from the mean.

Interpretation:

- $Z = 0 \rightarrow$ At mean
- $Z > 0 \rightarrow$ Above mean
- $Z < 0 \rightarrow$ Below mean

11. Difference Between PDF and CDF**PDF (Probability Density Function)**

Gives probability density

Used for continuous variables

Area under curve = Probability

CDF (Cumulative Distribution Function)

Gives cumulative probability

Used for both discrete and continuous variables

Value ranges from 0 to 1

Derivative of CDF

Integral of PDF